

Blood Sugar Regulation · Diabetes · Tests for Carbohydrate Disturbance

Syllabus lines 1–2

PART 1 / 5

01 HORMONAL CONTROL OF CARBOHYDRATE METABOLISM & REGULATION OF BLOOD SUGAR

Blood glucose is held within narrow limits by **one hypoglycaemic hormone opposed by five hyperglycaemic ones**. Sources of blood glucose: intestinal absorption, hepatic glycogenolysis, gluconeogenesis. Losses: tissue uptake and oxidation, glycogenesis, lipogenesis, and — above the renal threshold — urinary loss.

HORMONE	SOURCE	CHIEF ACTIONS ON CARBOHYDRATE	NET EFFECT
Insulin	β -cells, islets of Langerhans	Recruits GLUT-4 in muscle and fat; $\uparrow$ glucokinase, glycogen synthase, PFK-1, PDH, acetyl-CoA carboxylase; $\downarrow$ glycogen phosphorylase, gluconeogenic enzymes, hormone-sensitive lipase	The only hypoglycaemic hormone ; anabolic; also drives K^+ into cells
Glucagon	α -cells	cAMP $\rightarrow$ PKA $\rightarrow$ hepatic glycogenolysis, gluconeogenesis, ketogenesis ; no action on muscle	Hyperglycaemic; the fasting hormone
Adrenaline	Adrenal medulla	Glycogenolysis in liver and muscle , lipolysis, $\downarrow$ insulin release (α_2)	Hyperglycaemic; "stress hyperglycaemia" — marked in the cat
Glucocorticoids	Adrenal cortex	$\uparrow$ gluconeogenesis and protein catabolism; peripheral insulin resistance	Hyperglycaemic; hyperadrenocorticism
Growth hormone	Anterior pituitary	Anti-insulin, $\uparrow$ lipolysis	Hyperglycaemic; acromegaly, and the progesterone-driven GH of the dioestrous bitch
Thyroxine	Thyroid	$\uparrow$ intestinal absorption and glycogenolysis	Hyperglycaemic

70–110

dog/cat, mg/dL

45–60

cattle & sheep

75–115

horse

85–150

poultry

180

renal threshold, dog

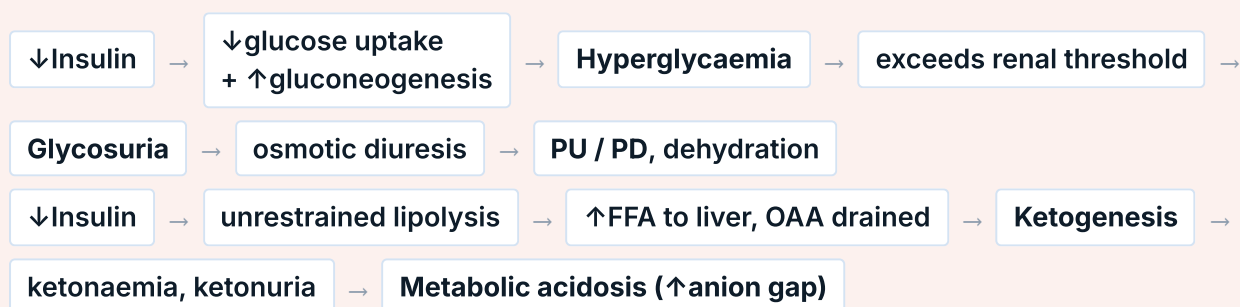
280–300

renal threshold, cat

Why ruminant values are low and why it matters: the ruminant absorbs almost no glucose from the gut and depends wholly on gluconeogenesis from **propionate**. Its "normal" 45–60 mg/dL is therefore **not** hypoglycaemia, and its margin for extra demand is thin — which is precisely why **ketosis and pregnancy toxemia are ruminant diseases and diabetes mellitus is rare in cattle**.

02 DIABETES MELLITUS & HYPERINSULINISM

Diabetes mellitus — chronic hyperglycaemia from absolute or relative insulin deficiency. **Type 1 (IDDM)**: β -cell destruction, absolute deficiency, ketosis-prone — the usual canine form. **Type 2 (NIDDM)**: insulin resistance with islet amyloid deposition — the usual feline form, linked to obesity and inactivity. **Secondary/type 3**: dioestrus and progesterone/GH in the bitch, hyperadrenocorticism, glucocorticoid or megestrol therapy, pancreatitis, acromegaly in the cat.



- **Cardinal signs — the "3 P's plus"**: polyuria, polydipsia, polyphagia with **weight loss**; later anorexia, vomiting, dehydration, acetone breath, **Kussmaul respiration**, cataract (dog — sorbitol/polyol pathway), plantigrade stance (cat — diabetic neuropathy), poor wound healing and recurrent infection, **hepatic lipidosis**.
- **Biochemical picture**: persistent fasting hyperglycaemia, glycosuria, ketonaemia/ketonuria, **metabolic acidosis with increased anion gap**, hypercholesterolaemia and lipaemia, $\uparrow$ ALP and ALT, dehydration with initial hyperkalaemia but **total-body potassium depletion**, and hyponatraemia (pseudo, from osmotic dilution).

Trap — the cat. Restraint alone can push a cat's glucose over 300 mg/dL by adrenaline release. **Stress hyperglycaemia is transient and does not cause a raised fructosamine**, so fructosamine (or glycated haemoglobin) is used to prove that hyperglycaemia has been present for weeks.

Hyperinsulinism (insulinoma) — a functional β -cell tumour, chiefly of the **middle-aged to older dog (ferrets also)**. Autonomous insulin secretion causes **fasting hypoglycaemia** with weakness, ataxia, muscle tremor, collapse and seizures, classically **relieved by feeding**.

Whipple's triad: neurological signs + documented hypoglycaemia + relief on giving glucose.

Diagnosis rests on **an inappropriately normal or high serum insulin measured on the same sample as a low glucose** (raised insulin : glucose ratio). Compare with the other causes of hypoglycaemia: hepatic failure and portosystemic shunt, hypoadrenocorticism, sepsis, neonatal and toy-breed hypoglycaemia, hunting-dog hypoglycaemia, xylitol ingestion, glycogen storage disease, and delay in separating serum (glycolysis in vitro).

03 BIOCHEMICAL TESTS FOR DISTURBANCE OF CARBOHYDRATE METABOLISM

TEST	PRINCIPLE	INTERPRETATION
Fasting blood glucose	GOD-POD (glucose oxidase–peroxidase) enzymatic method; or o-toluidine, Folin–Wu, Nelson–Somogyi (older)	The primary screen. Collect in NaF-oxalate (grey top) or separate serum within 30 min
Urine sugar	Benedict's (reducing sugars, non-specific) vs glucose-oxidase strip (specific for glucose)	Glycosuria = blood glucose above the renal threshold, or a proximal tubular defect
Ketone bodies	Rothera's nitroprusside test / dipstick — detects acetoacetate and acetone but NOT β-hydroxybutyrate	Ketonuria/ketonaemia. In serious ketosis most of the ketone is BHB, so the test underestimates severity
Serum / milk BHB	BHB dehydrogenase, hand-held meter; milk BHB test strips cow-side	The reference analyte for bovine ketosis: <1.0 normal, 1.2–1.4 subclinical, >2.6–3.0 clinical
Glucose tolerance test (GTT)	Oral or intravenous glucose load, then serial sampling	Diabetic curve: high fasting value, exaggerated peak, delayed return to baseline . Rarely needed and unreliable in the cat
Insulin tolerance / insulin : glucose ratio	Response to exogenous insulin; paired insulin and glucose	Insulin resistance; confirmation of insulinoma
Fructosamine	Glycated serum proteins — reflects the previous 1–3 weeks	Separates true diabetes from stress hyperglycaemia ; monitors control
Glycated Hb (HbA1c)	Glycation of haemoglobin — reflects 2–3 months	Long-term glycaemic control
Ancillary	Urine specific gravity, serum electrolytes, blood gas & anion gap, serum lipids, ALT/ALP, urine culture	Assess ketoacidosis, dehydration and complications

Guaranteed short note: "Benedict's test" — alkaline cupric citrate is reduced by a free carbonyl group; the sequence of colours **blue → green → yellow → orange → brick-red** grades roughly 0.5 %, 1 %, 1.5 % and 2 % of reducing sugar. It is positive for **any** reducing sugar (lactose in the lactating animal, fructose, galactose, and reducing drug metabolites), which is exactly why a **glucose-oxidase strip** is needed to prove that the sugar is glucose.

Ruminant & Neonatal Energy Disorders · Plasma Proteins

Bovine ketosis · pregnancy toxaemia · baby-pig hypoglycaemia · dysproteinaemias · acute-phase proteins

04 THE THREE CLASSICAL PRODUCTION-ANIMAL ENERGY DISORDERS — ONE MECHANISM, THREE SETTINGS

Glucose demand exceeds gluconeogenic supply → Hypoglycaemia →

↑glucagon, ↓insulin → lipolysis → NEFA flood the liver →

OAA diverted to gluconeogenesis → Acetyl-CoA → ketone bodies →

Ketonaemia · ketonuria · fatty liver

FEATURE	BOVINE KETOSIS (ACETONAEMIA)	PREGNANCY TOXAEMIA	BABY-PIG HYPOGLYCAEMIA
Species & timing	High-yielding dairy cow, first 2–6 weeks of lactation	Ewe or doe in the last 3–4 weeks of gestation, carrying twins or triplets; also seen in beef cows and queens	Piglet in the first 24–72 hours of life; also weak neonatal lambs, kids, kittens
Precipitating cause	Negative energy balance at peak yield; over-fat at calving; sudden ration change; displaced abomasum	Uterine glucose drain by multiple foetuses + reduced rumen capacity + any stress or transport	Negligible glycogen store, almost no body fat, immature gluconeogenesis, plus chilling, agalactia of the sow, large litter, splay leg
Biochemistry	↓glucose (<40 mg/dL) · ↑↑BHB >2.6–3.0 mmol/L · ketonuria · ↑NEFA · low hepatic glycogen · fatty liver	Same, but more severe and often irreversible : profound hypoglycaemia, massive ketonaemia, fatty liver, azotaemia, hypocalcaemia , terminal acidosis	Blood glucose <40 mg/dL, may fall below 20; ketosis is minimal because the piglet has too little fat to mobilise
Clinical signs	Drop in milk, selective inappetence (refuses concentrate), weight loss, firm dry dung, acetone smell in breath and milk; nervous form — licking, circling, bellowing, hyperaesthesia	Separation from flock, blindness, star-gazing, teeth grinding, incoordination, recumbency, coma; death in 2–7 days if untreated	Shivering, weakness, wobbling, hypothermia , high-pitched squealing, paddling convulsions, coma
Field tests	Rothera's test on urine/milk; cow-side milk or blood BHB strips ; Ross test	Urine ketones, blood glucose & BHB, ketones in aqueous humour post-mortem	Blood glucose; rule out hypothermia, starvation, septicaemia and iron-deficiency anaemia

Rational treatment	IV 40–50 % dextrose + propylene glycol / glycerol orally (gluconeogenic precursors) + glucocorticoids + B ₁₂ /cobalt; propionate salts; correct the ration	Same, plus induction of parturition or caesarean — removing the foetal glucose drain is the only definitive cure ; calcium borogluconate	Warmth first , then intraperitoneal or oral glucose every few hours; ensure colostrum intake and treat the sow's agalactia
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The examiner's favourite comparison: ketosis is a disease of the lactating cow with an outflow of glucose into milk; pregnancy toxemia is a disease of the pregnant ewe with an outflow into foetuses; baby-pig hypoglycaemia is a disease of absent reserve and absent intake. The first two are ketotic, the third is not — because ketosis requires fat to burn, and the newborn piglet has almost none ($\approx 1\text{--}2\%$ body fat and no brown adipose tissue).

Trap: ketosis has a **primary** form (pure negative energy balance) and a far commoner **secondary** form driven by another disease that stops the cow eating — **displaced abomasum, traumatic reticuloperitonitis, metritis, mastitis, lameness**. Never treat the ketones without hunting for the primary lesion.

05 PLASMA PROTEINS & DYSPROTEINAEMIAS

All plasma proteins except the immunoglobulins are made in the liver. On **cellulose-acetate electrophoresis at pH 8.6** they separate — fastest to slowest — as **albumin** $\rightarrow \alpha_1 \rightarrow \alpha_2 \rightarrow \beta \rightarrow \gamma$.

FRACTION	CHIEF MEMBERS	FUNCTION & DIAGNOSTIC MEANING
Albumin	Albumin (the single largest fraction)	$\approx 75\text{--}80\%$ of colloid osmotic pressure ; transports bilirubin, free fatty acids, calcium, hormones and drugs; a negative acute-phase protein ; the body's protein reserve
α_1	α_1 -antitrypsin, α_1 -acid glycoprotein, HDL	Antiprotease, acute-phase
α_2	Haptoglobin , ceruloplasmin, α_2 -macroglobulin	Haptoglobin binds free haemoglobin — falls in intravascular haemolysis ; ceruloplasmin carries copper
β	Transferrin , complement C3, fibrinogen, β -lipoprotein, IgM	Iron transport, complement
γ	Immunoglobulins IgG, IgA, IgM, IgE	Humoral immunity; made by plasma cells, not by the liver

6–8

total protein, g/dL

2.5–4.5

albumin, g/dL

0.6–1.2

A : G ratio, cattle

0.7–1.1

A : G ratio, dog

- **Hypoalbuminaemia** — reduced synthesis (chronic liver disease, malnutrition, chronic inflammation), increased loss (**protein-losing nephropathy — glomerulonephritis, amyloidosis; protein-losing enteropathy; burns; severe haemorrhage**), or sequestration. Consequences: **oedema, ascites and effusions** once albumin falls below ≈ 1.5 g/dL — the direct application of oncotic pressure and the Donnan effect from Unit 1.
- **Hyperglobulinaemia** — **polyclonal** (a broad-based γ peak: chronic infection, ehrlichiosis, FIP, trypanosomiasis, dirofilariasis, chronic dermatitis, liver disease) or **monoclonal** (a tall narrow spike: **multiple myeloma, plasmacytoma, lymphoma, monoclonal gammopathy**).
- **A : G ratio** falls in almost every chronic disease — either because albumin falls, globulin rises, or both. A total protein that looks normal can hide a severe dysproteinaemia, which is why **the ratio, not the total, is the diagnostic figure**.
- **Failure of passive transfer** in the calf, foal, lamb and kid — inadequate colostral IgG gives a low total protein and low γ fraction in the first days of life; screened by **total protein by refractometer, zinc sulphate turbidity, glutaraldehyde coagulation or GGT**.

06 ACUTE-PHASE PROTEINS (APPS)

Definition: plasma proteins whose concentration changes by at least 25 % within 24–48 hours of tissue injury, infection, inflammation, surgery or neoplasia, driven by **IL-1, IL-6 and TNF- α** acting on the hepatocyte. They are **non-specific but exquisitely sensitive early markers**, and they rise long before leucocyte changes appear.

DIRECTION	PROTEIN	SPECIES WHERE IT IS THE MAJOR APP
Positive (rise)	Haptoglobin	Cattle, sheep, goat — the classic bovine marker (mastitis, metritis, pneumonia, traumatic reticulitis)
	Serum amyloid A (SAA)	Horse and cat ; fastest responder of all — rises within 6–8 h
	C-reactive protein (CRP)	Dog and pig
	α_1 -acid glycoprotein, fibrinogen, ceruloplasmin, complement, α_1 -antitrypsin	Fibrinogen is the practical, cheap ruminant and equine inflammation marker
Negative (fall)	Albumin , transferrin, transthyretin	All species — synthesis is switched away from albumin toward APPs

Species shortcut — commit this to memory: Cattle → Haptoglobin · Horse & Cat → SAA · Dog & Pig → CRP. Fibrinogen and the plasma-fibrinogen : total-protein ratio remain the cheapest field test in large animals.

Uses: early detection of subclinical disease in herds (transition-cow monitoring, pig-herd health), differentiating inflammatory from non-inflammatory disease, monitoring response to therapy, prognosis in surgery and colic, and **meat-inspection screening**. Chronic persistent elevation of SAA underlies **reactive (AA) amyloidosis**.

07 LIPID PROFILE IN DISEASE DIAGNOSIS

The panel: **total cholesterol, triacylglycerol, HDL-cholesterol, LDL-cholesterol** (Friedewald estimate), phospholipids and NEFA. A fasting sample of 12 h is essential — post-prandial chylomicrons make the serum **lipaemic (milky)** and interfere with almost every other assay.

CHANGE	INTERPRET AS
Hyper-cholesterolaemia	Hypothyroidism (the classic canine cause) , diabetes mellitus, hyperadrenocorticism, nephrotic syndrome , cholestasis/post-hepatic obstruction, pancreatitis, high-fat diet, idiopathic hyperlipidaemia of the Miniature Schnauzer and Shetland Sheepdog
Hypo-cholesterolaemia	Hepatic insufficiency and portosystemic shunt , protein-losing enteropathy, maldigestion/malabsorption, starvation, hypoadrenocorticism
Hyper-triglyceridaemia	Post-prandial, diabetes mellitus, pancreatitis, hypothyroidism, hyperadrenocorticism, cholestasis, equine hyperlipaemia
↑ NEFA	Negative energy balance — the transition dairy cow; predicts ketosis and fatty liver

Two species syndromes to name: ① **Equine hyperlipaemia** — obese **ponies, miniature horses and donkeys**, especially pregnant or lactating mares under stress or anorexia; massive VLDL/TG rise, opaque plasma, fatty infiltration of liver and kidney, high mortality. ② **Feline hepatic lipidosis** — the anorectic obese cat mobilises fat faster than its liver can export VLDL. Both illustrate the same principle: **fat mobilisation outruns hepatic export capacity**.

Trap: dogs and cats are "**HDL animals**" — HDL, not LDL, is the predominant lipoprotein — so **spontaneous atherosclerosis is rare**. A hypercholesterolaemic dog is telling you about its thyroid, adrenal, kidney or pancreas, not about its coronary arteries.

08 CLINICAL ENZYMOLOGY

Functional plasma enzymes are secreted into plasma to act there and rise in liver disease (coagulation factors, lipoprotein lipase, ceruloplasmin, pseudocholinesterase — these **fall** in hepatic failure). **Non-functional plasma enzymes** have no role in plasma; they leak from damaged cells and their **rise indicates cell injury or induced synthesis** — these are the diagnostic enzymes.

ENZYME	CHIEF TISSUE	RISES IN	SPECIES NOTE
ALT (SGPT)	Hepatocyte cytosol	Hepatocellular damage — the liver-specific leakage enzyme	Useful only in dog, cat and primate; of no value in cattle, sheep, horse

AST (SGOT)	Liver + muscle + RBC	Hepatocellular and muscle damage; haemolysis	The liver enzyme of large animals , always paired with CK to decide liver vs muscle
SDH / GLDH	Hepatocyte	Acute hepatocellular necrosis	Liver-specific in horses and ruminants ; SDH is very labile
ALP	Biliary epithelium, bone, intestine	Cholestasis ; bone growth; drugs	Steroid-induced isoenzyme is unique to the dog ; short half-life in the cat makes any rise significant
GGT	Biliary epithelium; colostrum	Cholestasis , especially in horse, ruminant and cat	Very high in colostrum — used to confirm passive transfer in calves and foals
CK (CPK)	Skeletal & cardiac muscle, brain	Acute muscle damage — the most specific and earliest muscle marker; short half-life	Equine exertional rhabdomyolysis, white muscle disease (Se/vit E), downer cow, IM injections
LDH	Ubiquitous	Any tissue damage; haemolysis	Sensitive but non-specific; isoenzymes localise it
Amylase & lipase	Pancreas (also salivary, gastric)	Acute pancreatitis in the dog	Also rise in renal failure (reduced clearance); DGGR-lipase / cPLI are more specific
Cholinesterase	Plasma & RBC	Falls — never rises	<25 % of normal confirms organophosphate/carbamate poisoning ; also falls in hepatic failure

Read the pattern, not the enzyme: ALT/AST/SDH ↑ = hepatocellular (leakage) · ALP/GGT ↑ = cholestatic (induction) · CK + AST ↑ but ALT normal = muscle · albumin ↓, urea ↓, glucose ↓, cholesterol ↓, bilirubin ↑ = loss of hepatic **function**. Damage and function are two different questions.

09 LIVER FUNCTION TESTS — CLASSIFICATION & DIFFERENTIAL DIAGNOSIS OF JAUNDICE

Classification of LFTs — the framework the examiner wants before any detail:

- ① **Excretory** **Bilirubin** (total, direct/conjugated, indirect), urine bilirubin and urobilinogen, **BSP (bromosulphthalein)** and **ICG clearance**, serum **bile acids** (pre- and 2 h post-prandial — the single most sensitive functional test, and the test of choice for a **portosystemic shunt**).
- ② **Enzymes of damage** **ALT, AST, SDH, GLDH** (hepatocellular) and **ALP, GGT** (cholestatic).
- ③ **Synthetic function** **Albumin, total protein, A : G ratio, cholesterol, blood urea, glucose, coagulation factors (PT, APTT)** — all **fall** in hepatic failure.

④ **Detoxifying function**

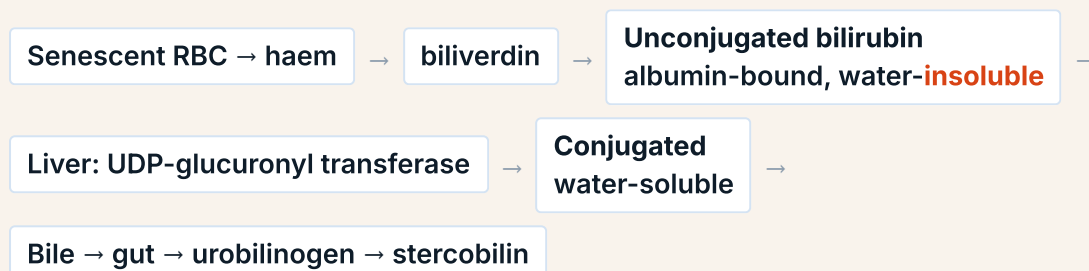
Blood ammonia, ammonia tolerance test, hippuric acid test.

⑤ **Protein-flocculation**

Older but still asked: **thymol turbidity**, **zinc sulphate turbidity**, **cephalin-cholesterol flocculation**, Takata-Ara.

Trap: the liver has an enormous reserve — **more than 70–80 % must be lost before the functional tests become abnormal**. Enzymes tell you about damage **early**; albumin, urea and bilirubin tell you about failure **late**.

Bilirubin metabolism in one line:



TEST	PRE-HEPATIC (HAEMOLYTIC)	HEPATIC	POST-HEPATIC (OBSTRUCTIVE)
Total bilirubin	↑	↑	↑↑↑
Predominant form	Unconjugated (indirect)	Both	Conjugated (direct)
van den Bergh	Indirect positive	Biphasic	Direct, prompt
Urine bilirubin	Absent	Present	Present ++
Urine urobilinogen	↑↑	Variable	Absent
Faeces	Dark	Normal/pale	Clay-coloured, pale
ALT / AST	Normal or mild ↑	↑↑↑	Mild ↑
ALP / GGT	Normal	Mild ↑	↑↑↑
PCV	↓ anaemia	Normal	Normal

This differential table is the single most repeated long-answer question in Unit 3. Reproduce it as a table, then add one line of species context: **the horse develops fasting (anorexia) hyperbilirubinaemia** that is unconjugated and physiological; **cattle rarely become icteric** except in haemolytic disease (babesiosis, anaplasmosis, leptospirosis, copper toxicity, **Brassica**); **any icterus in a cat demands exclusion of hepatic lipidosis, cholangiohepatitis, FIP and haemoplasmosis**.

Common causes to quote: **pre-hepatic** — babesiosis, theileriosis, anaplasmosis, trypanosomiasis, leptospirosis, immune-mediated haemolytic anaemia, onion/copper/paracetamol toxicity, incompatible transfusion. **Hepatic** — infectious canine hepatitis, leptospirosis, aflatoxicosis and other mycotoxins, **lantana and Crotalaria/pyrrolizidine alkaloid poisoning**, copper toxicity in sheep, chronic fibrosis, neoplasia, drugs. **Post-hepatic** — cholelithiasis, **Fasciola in ruminants**, ascariasis in the pig, pancreatitis, neoplasia of the duct or pancreas.

Ammonia and encephalopathy: a failing or bypassed liver cannot run the urea cycle, so **blood ammonia rises while BUN falls**. Together with **ammonium biurate crystals** in the urine and abnormal **bile acids**, this is the diagnostic triad of a **portosystemic shunt** in a young dog.

Renal Function Tests · Urine Analysis · Acid–Base Disorders

Syllabus lines 8–9

PART 4 / 5

10 BIOCHEMICAL TESTS OF RENAL FUNCTION

ANALYTE	ORIGIN & BEHAVIOUR	INTERPRETATION
BUN / blood urea	End product of the hepatic urea cycle ; freely filtered but partly reabsorbed , and reabsorption rises when flow is slow	↑ pre-renal (dehydration, shock, cardiac failure, high-protein diet, GI haemorrhage, catabolism), renal ($\geq 75\%$ nephron loss) or post-renal (obstruction, ruptured bladder). ↓ in hepatic failure, portosystemic shunt, low-protein diet, over-hydration, and in ruminants, in whom urea is recycled to the rumen
Creatinine	From muscle creatine phosphate at a constant daily rate ; freely filtered and not reabsorbed	The better index of GFR. Independent of diet and protein catabolism, but influenced by muscle mass — low in cachectic and very young animals. Rises only after $\approx 75\%$ of nephrons are lost
Uric acid	End product of purine catabolism	Chief nitrogenous end product in birds and reptiles — raised in avian gout and renal disease; in the dog it matters mainly in the Dalmatian (urate uroliths) and in portosystemic shunt
Clearance tests	Inulin clearance = gold standard for GFR ; endogenous or exogenous creatinine clearance is the practical version; PAH clearance measures renal plasma flow & tubular secretion	Detect renal disease before azotaemia appears; $C = (U \times V)/P$
SDMA	Symmetric dimethylarginine, freely excreted	Rises with only $\sim 25\text{--}40\%$ loss of GFR — the modern early marker
Others	Serum phosphorus & calcium, potassium, sodium, bicarbonate, urine protein : creatinine ratio , water deprivation & concentration test, fractional electrolyte excretion, GGT : creatinine ratio in urine (tubular damage)	Stage the disease and separate glomerular from tubular injury

Azotaemia vs uraemia: azotaemia is a **laboratory** finding — raised blood urea and creatinine. **Uraemia** is the **clinical syndrome** — azotaemia plus vomiting, ulcerative stomatitis, uraemic breath, anaemia, acidosis. The **BUN : creatinine ratio** and the **urine specific gravity taken before fluid therapy** are what separate pre-renal from renal azotaemia: a dehydrated animal with pre-renal azotaemia still concentrates its urine (USG >1.030 dog, >1.035 cat), a renal-failure patient cannot (isosthenuria 1.008–1.012).

11 URINE ANALYSIS — DETECTION OF PATHOLOGICAL CONSTITUENTS

FINDING	TEST	MEANS
Glucose	Benedict's; glucose-oxidase strip	Diabetes mellitus; also proximal tubular (Fanconi) defect, stress in the cat
Ketones	Rothera's nitroprusside ; strip	Ketosis, pregnancy toxemia, diabetic ketoacidosis, starvation. Does not detect BHB
Protein	Heat & acetic acid, sulphosalicylic acid, dipstick; UPC ratio to quantify	Glomerulonephritis, amyloidosis, pyelonephritis, cystitis, haemorrhage; must be read with the USG and sediment
Bilirubin	Fouchet's , Gmelin's, strip	Hepatic and post-hepatic jaundice. Trace bilirubinuria is normal in the dog (low renal threshold) but never normal in the cat
Urobilinogen	Ehrlich's aldehyde	Increased in haemolysis; absent in complete biliary obstruction
Blood / Hb / Mb	Benzidine or o-toluidine; sediment	Intact RBC = haematuria; free Hb = haemoglobinuria (babesiosis, leptospirosis, post-parturient haemoglobinuria); myoglobin = rhabdomyolysis, azoturia, white muscle disease
Casts & cells	Sediment microscopy	Casts are moulded in the tubule and prove renal, not lower-tract, origin : hyaline, granular, cellular, waxy
Crystals	Sediment	Struvite, calcium oxalate, ammonium biurate (portosystemic shunt / Dalmatian) , cystine, urate; calcium oxalate monohydrate in ethylene-glycol poisoning
Physical	Volume, colour, transparency, odour, specific gravity by refractometer , pH	Isosthenuria 1.008–1.012 = the kidney has lost concentrating ability . Herbivore urine is normally alkaline, carnivore acidic

Guaranteed practical question: the sequence of a complete urinalysis — **physical → chemical → microscopic** — and why the sample must be fresh (within 30 min or refrigerated): standing urine loses ketones and bilirubin, grows bacteria, turns alkaline as urea is split to ammonia, and dissolves or precipitates crystals and casts.

12 DISTURBANCES OF ACID-BASE BALANCE AND THEIR DIAGNOSIS

Everything follows from **Henderson–Hasselbalch**: $\text{pH} = 6.1 + \log (\text{HCO}_3^- / 0.03 \times \text{pCO}_2)$, normally $6.1 + \log 20 = 7.4$. HCO_3^- is the **metabolic (renal) component**; pCO_2 is the **respiratory (pulmonary) component**. Name the primary disorder by asking which one moved in the same direction as the pH change.

DISORDER	PH	PRIMARY CHANGE	COMPENSATION	VETERINARY CAUSES
Metabolic acidosis	↓	HCO_3^- ↓	Hyperventilation → pCO_2 ↓ (fast, hours)	Diabetic ketoacidosis, ruminal lactic acidosis (grain overload), diarrhoea in calves (HCO_3^- loss), renal failure, shock & lactic acidosis, ethylene glycol, urinary obstruction
Metabolic alkalosis	↑	HCO_3^- ↑	Hypoventilation → pCO_2 ↑ (limited by hypoxia)	Abomasal displacement/torsion and vomiting (loss of HCl), high-dose loop diuretics, hypokalaemia, excess bicarbonate therapy
Respiratory acidosis	↓	pCO_2 ↑	Renal H^+ excretion & HCO_3^- retention (slow, 2–5 days)	Airway obstruction, pneumonia, pulmonary oedema, anaesthetic depression, upper-airway paralysis, pleural effusion
Respiratory alkalosis	↑	pCO_2 ↓	Renal HCO_3^- excretion	Heat stress and panting in poultry and cattle — the classic cause of thin-shelled eggs ; pain, fever, hyperventilation, early sepsis

Anion gap = $(\text{Na}^+ + \text{K}^+) - (\text{Cl}^- + \text{HCO}_3^-)$, normally $\approx 12\text{--}25$ mEq/L. A **high anion gap** means unmeasured acid has been added — **ketoacids, lactate, uraemic acids, ethylene-glycol metabolites**. A **normal (hyperchloraemic) anion gap** means bicarbonate has simply been lost — **diarrhoea, renal tubular acidosis**. This one calculation separates the two families of metabolic acidosis.

How to read a blood-gas report in four steps: ① look at **pH** — acidaemia or alkalaemia; ② look at **HCO_3^- and pCO_2** and see which one explains the pH — that is the primary disorder; ③ check whether the other has moved in the **same direction** — that is compensation (compensation never overshoots to normal); ④ if both move in opposite directions, or the compensation is inappropriate, call it a **mixed disorder**.

7.35–7.45

arterial pH

35–45

pCO_2 , mmHg

22–26

HCO_3^- , mmol/L

±2

base excess

12–25

anion gap, mEq/L

Two field syndromes to be able to write out in full: ① **Ruminal lactic acidosis (grain overload)** — sudden carbohydrate load → ***Streptococcus bovis*** and lactobacilli produce **D- and L-lactate** → **rumen pH falls below 5** → chemical rumenitis, osmotic pull of water into the rumen, dehydration, **systemic metabolic acidosis**, laminitis and hepatic abscessation. Treat with oral alkalinisers (magnesium hydroxide), rumenotomy or transfaunation, and **IV sodium bicarbonate with fluids**. ② **Calf diarrhoea** — loss of bicarbonate and sodium in faeces plus D-lactate absorption → **metabolic acidosis with dehydration and paradoxical hyperkalaemia**; the whole rationale of oral rehydration salts containing bicarbonate/citrate precursors.

Trap: acidosis drives **K⁺ out of cells** (in exchange for H⁺) so serum potassium looks **high** even though total body potassium is depleted; correcting the acidosis with bicarbonate and insulin then drives K⁺ back in and can precipitate fatal **hypokalaemia**. Always supplement potassium once the animal is urinating.

Digestive Disorders · Oxidative Stress & Shock · Fluid Therapy · Detoxification

Syllabus lines 10–12, plus the final recall bank

PART 5 / 5

13 BIOCHEMISTRY OF DIGESTIVE DISORDERS

- **Maldigestion vs malabsorption.** **Maldigestion** = failure to break food down — chiefly **exocrine pancreatic insufficiency (EPI)**, common in the **German Shepherd**: steatorrhea, voluminous pale faeces, ravenous appetite with weight loss; confirmed by a **low serum TLI (trypsin-like immunoreactivity)**, with low cobalamin and high folate. **Malabsorption** = failure of the mucosa to take up the products — infiltrative bowel disease, villous atrophy, parasitism, lymphangiectasia; assessed by **D-xylose absorption, faecal fat, cobalamin and folate**.
- **Protein-losing enteropathy** — loss of albumin **and** globulin into the gut (lymphangiectasia, chronic enteropathy, parasitism, Johne's disease) → **panhypoproteinaemia, oedema, ascites**. Contrast with protein-losing **nephropathy**, which loses albumin selectively.
- **Ruminal disorders — the biochemical trio: simple indigestion** (rumen pH 6–7, poor motility); **ruminal lactic acidosis** (pH <5, ↑D- and L-lactate, systemic acidosis); **ruminal alkalosis** from urea overfeeding or putrefaction (**pH >7.5–8, high ammonia**). Rumen fluid examination — colour, odour, consistency, **pH, methylene-blue reduction time, protozoal motility, sedimentation-flotation, chloride (>30 mEq/L suggests abomasal reflux)** — is a favourite practical question.
- **Abomasal displacement and vomiting** sequester HCl → **hypochloraemic, hypokalaemic metabolic alkalosis with paradoxical aciduria**.
- **Diarrhoea** — secretory (enterotoxigenic **E. coli**, cholera-like, mediated by cAMP), osmotic (lactose intolerance, overfeeding), inflammatory/exudative, and dysmotility. All lose **water, sodium, potassium and bicarbonate** → dehydration + metabolic acidosis.

14 OXIDATIVE STRESS & SHOCK

Oxidative stress = an imbalance between the production of **reactive oxygen species (ROS)** and the antioxidant defence, in favour of the ROS, causing damage to lipid, protein and DNA.



H₂O

- **Sources of ROS:** leakage from the electron transport chain, the **respiratory burst** of neutrophils (NADPH oxidase, myeloperoxidase), xanthine oxidase during **ischaemia-reperfusion**, cytochrome P450, transition metals via the **Fenton and Haber–Weiss reactions**, radiation, mycotoxins and xenobiotics.

- **Antioxidant defence — enzymatic:** superoxide dismutase (Cu-Zn cytosolic, Mn mitochondrial), catalase, glutathione peroxidase (selenium), glutathione reductase (needs NADPH from the HMP shunt), thioredoxin. **Non-enzymatic:** vitamin E (α -tocopherol — the chain-breaking membrane antioxidant), vitamin C, β -carotene, glutathione, uric acid, taurine, ceruloplasmin, transferrin, albumin, zinc.
- **Consequences:** lipid peroxidation (measured as **MDA / TBARS**), membrane and enzyme damage, DNA strand breaks, and activation of NF- κ B with cytokine release.
- **Veterinary conditions in which oxidative stress is central:** white muscle disease and mulberry heart disease (Se/vitamin E deficiency), yellow fat disease in cats, transition-cow disease and retained placenta, mastitis and metritis, ischaemia–reperfusion after gastric dilatation–volvulus and colic, **onion, copper, paracetamol, nitrate and mycotoxin poisonings**, heat stress in poultry, and ageing.

Shock — acute generalised failure of tissue perfusion. Whatever the type (hypovolaemic, distributive/septic, cardiogenic, obstructive), the biochemistry converges: **inadequate O₂ delivery → anaerobic glycolysis → lactic acid accumulation → metabolic acidosis with a high anion gap**; the Na⁺/K⁺-ATPase fails for want of ATP so **cells swell and K⁺ leaks out**; lysosomal enzymes are released; **endotoxin (lipid A)** in septic shock triggers TNF- α , IL-1 and nitric oxide, causing vasodilation, capillary leak and **DIC**. Monitoring: **blood lactate (the best prognostic indicator)**, base deficit, PCV/TP, glucose, electrolytes, creatinine, coagulation profile.

15 BIOCHEMICAL BASIS OF FLUID THERAPY • DETOXIFICATION AND XENOBIOTIC METABOLISM

Fluid compartments. Total body water $\approx$ **60 % of body weight**, split as **ICF $\approx$ 40 %** and **ECF $\approx$ 20 %** (interstitial 15 %, plasma 5 %). **Na⁺ is the principal ECF cation and the determinant of ECF volume; K⁺ is the principal ICF cation.** Water moves by osmosis; protein holds it in the vessel by oncotic pressure (Starling forces + the Donnan effect).

FLUID	TYPE	USE
Normal saline 0.9 %	Isotonic crystalloid, replacement	Hypochloraemic alkalosis, hyperkalaemia, hypercalcaemia ; contains no buffer
Ringer's lactate / acetate	Balanced isotonic crystalloid	The default replacement fluid ; lactate is metabolised in the liver to bicarbonate — so use acetate when the liver is failing
5 % dextrose	Free water once metabolised	Pure water deficits, hypoglycaemia, ketosis (40–50 % IV) ; never for volume replacement
Sodium bicarbonate	Alkalinising	Severe metabolic acidosis — grain overload, calf diarrhoea, ketoacidosis. Deficit $\approx$ BW(kg) $\times$ 0.3 $\times$ base deficit
Hypertonic saline 7.2 %	Small-volume resuscitation	Shock, GDV; must be followed by isotonic fluid
Colloids & plasma	Oncotic support	Hypoalbuminaemia, severe hypoproteinaemia

Prescribing in three parts: ① **correct the existing deficit** = $\text{BW(kg)} \times \% \text{ dehydration} \times 10 = \text{mL}$ · ② **replace ongoing losses** (vomiting, diarrhoea, polyuria) · ③ **supply maintenance** $\approx 50\text{--}60 \text{ mL/kg/day}$. Then decide the **route**, the **rate**, and the **potassium** — never exceed $0.5 \text{ mEq K}^+/\text{kg/hour}$.

Detoxification (biotransformation) converts lipid-soluble xenobiotics — drugs, plant toxins, pesticides, mycotoxins, food additives, industrial chemicals — into **more water-soluble, less toxic, more readily excretable** compounds. The **liver** is the chief site (smooth ER); also kidney, lung, intestinal mucosa and skin.

Phase I — functionalisation **Oxidation, reduction, hydrolysis**, hydration and dehalogenation. Introduces or unmask a polar group ($-\text{OH}$, $-\text{NH}_2$, $-\text{SH}$, $-\text{COOH}$). Dominated by the **cytochrome P450 mixed-function oxidase system**. Products may be **more** toxic than the parent — **bioactivation**.

Phase II — conjugation Attachment of a bulky polar endogenous molecule, always making the product **less toxic and readily excreted**: **glucuronidation (UDP-glucuronyl transferase — the commonest)**, **sulphation (PAPS)**, **glutathione conjugation**, **acetylation**, **methylation (SAM)**, **glycine conjugation (hippuric acid)**, and **mercapturic-acid formation**.

Cytochrome P450 — a superfamily of **haem-thiolate monooxygenases** named for the 450 nm absorbance of their CO complex. Reaction: $\text{RH} + \text{O}_2 + \text{NADPH} + \text{H}^+ \rightarrow \text{ROH} + \text{H}_2\text{O} + \text{NADP}^+$ — **one atom of oxygen enters the substrate, the other becomes water**; the partner enzyme is **NADPH-cytochrome P450 reductase**. Besides xenobiotics, P450s make **steroid hormones, bile acids, vitamin D and eicosanoids**. They are strongly **inducible** (phenobarbitone, rifampicin, PCBs, some pesticides) and **inhibitable** (chloramphenicol, cimetidine, ketoconazole) — the mechanism of countless drug interactions and of altered withdrawal periods in food animals.

Species deficiencies — the most rewarding facts in this section: ① **The cat has a deficient glucuronyl transferase**, so it cannot glucuronidate efficiently — hence its notorious sensitivity to **paracetamol (methaemoglobinaemia, Heinz bodies, facial oedema), aspirin, phenols and benzoic acid**. ② **The dog is a poor acetylator** (deficient N-acetyltransferase) — sulphonamide handling differs. ③ **The pig sulphates poorly**. ④ Neonates of all species have immature phase-II systems. Antidote logic follows directly: **N-acetylcysteine replenishes glutathione** in paracetamol poisoning.

Trap — bioactivation. "Detoxification" does not always detoxify. Phase I can create a more dangerous metabolite: **paracetamol** → **NAPQI** (hepatotoxic and, in cats, met-Hb forming), **parathion** → **paraoxon** (a far stronger cholinesterase inhibitor), **aflatoxin B₁** → **its 8,9-epoxide** (binds DNA, hepatocarcinogenic), **ethylene glycol** → **glycolic and oxalic acid** (metabolic acidosis and calcium-oxalate nephrosis, treated by blocking alcohol dehydrogenase with fomepizole or ethanol). Always name one such example when asked about biotransformation.

16 FINAL RECALL BANK — READ THESE LAST, STRAIGHT BEFORE THE PAPER

Only hypoglycaemic hormone	Insulin
Insulin-dependent glucose transporter	GLUT-4
Normal blood glucose, cattle	45–60 mg/dL
Renal threshold for glucose, dog / cat	≈180 / ≈280 mg/dL
Anticoagulant for a glucose sample	Sodium fluoride–oxalate
Test that distinguishes stress from true diabetes	Fructosamine
Ketone body not detected by Rothera's	β-hydroxybutyrate
Diagnostic analyte for bovine ketosis	Blood or milk BHB
Clinical BHB cut-off	>2.6–3.0 mmol/L
Ketosis occurs in which stage of lactation	First 2–6 weeks (peak yield)
Pregnancy toxæmia occurs when	Last 3–4 weeks, twins/triplets
Definitive cure of pregnancy toxæmia	Remove the foetuses
Why baby pigs are not ketotic	Almost no body fat to mobilise
Oral gluconeogenic precursor for ketosis	Propylene glycol / glycerol
Functional β-cell tumour	Insulinoma — Whipple's triad
Electrophoretic order of plasma proteins	Albumin, α₁, α₂, β, γ
Protein contributing most oncotic pressure	Albumin (75–80 %)
Only plasma proteins not made in liver	Immunoglobulins
Protein binding free haemoglobin	Haptoglobin
Tall narrow γ spike means	Monoclonal gammopathy / myeloma
Major acute-phase protein in cattle	Haptoglobin
Major APP in horse and cat	Serum amyloid A
Major APP in dog and pig	C-reactive protein
The negative acute-phase protein	Albumin
Cytokines driving the APP response	IL-1, IL-6, TNF-α
Classic cause of canine hypercholesterolaemia	Hypothyroidism
Hyperlipaemia of ponies and donkeys	Equine hyperlipaemia
Dogs and cats are which lipoprotein animals	HDL animals
Liver leakage enzyme in dog and cat	ALT
Liver enzyme of large animals	AST (with SDH / GLDH)
Enzymes of cholestasis	ALP and GGT

Best marker of acute muscle damage	Creatine kinase
Enzyme used to confirm passive transfer	GGT (high in colostrum)
Enzyme that falls in OP poisoning	Cholinesterase (<25 %)
Most sensitive functional liver test	Serum bile acids
Urine bilirubin in haemolytic jaundice	Absent
Urine urobilinogen in obstruction	Absent
Direct van den Bergh means	Conjugated / obstructive
Portosystemic shunt triad	↑ammonia, ↓BUN, biurate crystals
% liver that must be lost before failure	>70–80 %
Better index of GFR than urea	Creatinine
Gold-standard GFR marker	Inulin clearance
% nephron loss before azotaemia	≈75 %
Urea is low in	Liver failure & portosystemic shunt
Nitrogenous end product of birds	Uric acid (uricotelic)
Isosthenuria range	1.008–1.012
Casts in urine prove	Renal (tubular) origin
Crystals in ethylene-glycol poisoning	Calcium oxalate monohydrate
Henderson–Hasselbalch ratio in blood	$\text{HCO}_3^- : \text{H}_2\text{CO}_3 = 20 : 1$
Metabolic component / respiratory component	$\text{HCO}_3^- / \text{pCO}_2$
Anion gap formula	$(\text{Na}+\text{K}) - (\text{Cl}+\text{HCO}_3)$
Acid–base state in grain overload	Metabolic acidosis (rumen pH <5)
Acid–base state in displaced abomasum	Hypochloraemic metabolic alkalosis
Acid–base state in panting poultry	Respiratory alkalosis
Test for exocrine pancreatic insufficiency	Serum TLI
Selenium-dependent antioxidant enzyme	Glutathione peroxidase
Marker of lipid peroxidation	MDA / TBARS
Best prognostic indicator in shock	Blood lactate
Default balanced replacement fluid	Ringer's lactate
Commonest phase-II conjugation	Glucuronidation
Conjugation reaction deficient in the cat	Glucuronidation
Toxic metabolite of paracetamol	NAPQI (antidote N-acetylcysteine)
Enzyme system of phase-I oxidation	Cytochrome P450 (mixed-function oxidase)

Thirty-second closing sweep: insulin alone lowers glucose; five hormones raise it · fructosamine beats stress hyperglycaemia · Rothera's misses BHB · ketosis = lactation, toxemia = twins, baby pig = no reserve · haptoglobin cattle, SAA horse/cat, CRP dog/pig, albumin negative · ALT/AST leak, ALP/GGT cholestasis, CK muscle, bile acids function · haemolytic = indirect + no urine bilirubin; obstructive = direct + no urobilinogen · creatinine beats urea; inulin is the gold standard; 75 % nephrons before azotaemia · HCO_3^- metabolic, pCO_2 respiratory, anion gap splits the acidoses · $\text{SOD} \rightarrow \text{catalase/GSH-Px(Se)} \rightarrow \text{water}$ · phase I P450 can bioactivate; phase II always detoxifies; the cat cannot glucuronidate.